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Requirements Prioritization Techniques Focusing on Agile Software Development: A Systematic Literature Review

Noor Hazlini Borhan, Hazura Zulzalil, Sa'adah Hassan, Norhayati Mohd Ali

Abstract— Requirement prioritization plays a crucial role in the software development process. In Agile Software Development (ASD), requirement prioritization (RP) is challenging to maintain and requires a more formal process. It is essential to prioritize the requirements for making the correct decision for either single or multiple releases of a product. Ignorance of critical requirements during prioritization results in poor quality and unsatisfied clients. Various techniques have been proposed to prioritize requirements based on specific criteria. This SLR aims to identify the RP techniques focusing on Agile Software Development, as well as the issues or limitations of previous works. It will serve as a guideline for the practitioners to develop an effective RP process that can produce high-quality requirements that are expected to meet the needs of the intended users. Search terms with appropriate keywords were utilized in several online databases to recognize primary studies mostly related to RP in ASD. This SLR also shows the significant research gaps regarding techniques and issues for requirements prioritization in ASD. The researchers identified that a total of 25 papers report complete empirical results. Some critical issues of RP in ASD include severe limitations in terms of scalability, complexity, uncertainty, time consumption, starvation issue, dependency issues among requirements, limited research focused on the non-functional requirements and a lower automation approach. There are some significant challenges of RP techniques in ASD, such as a conflict between the stakeholders, changes in the priority list lead towards rework, and requirement selection factors during the RP process.

Index Terms— Requirements Prioritization, Prioritization Techniques, Prioritization Factors, Agile Software Development, Systematic Review

1 INTRODUCTION

Requirement Prioritization is a primary phase in requirement engineering, as not all requirements have equivalent customer fulfillment. From the viewpoint of the project manager, the requirements that can be implemented easily should be implemented first [1]. Conversely, the financial manager is in favor of the requirements that cost less [1]. Each requirement has different interests depending on the software to be built. Many factors affect the importance of each requirement. Delivering functionalities which achieve customer satisfaction and fit into the project constraints, while discarding requirements that are not worth implementing is the aim of requirement prioritization [2]. RP plays a role in the success or failure of a software product. If the correct or essential requirements are not prioritized at the right time, a software product can go wrong or fail. Hence, there is a need for an endless or complete requirements prioritization technique, model, or framework that extends all the factors that must be considered while prioritizing requirements in ASD.

The success of quality software development depends on the correct choice of requirement candidates, which are then prioritized based on essential priority perspectives. Various techniques are used in prioritizing requirements. Based on a reviewed done in this study, these techniques include the Analytic Hierarchy Process (AHP), Value-Oriented Prioritization (VOP), Wieger's method, MoSCoW technique, Hierarchical Cumulative Voting (HCV), Goal-Based Technique, Planning Game (PG) and Agile User Stories Prioritization (AUSP) method.

Additionally, the prioritizing of requirements may use hybrid techniques such as Adaptive Fuzzy Hierarchical Cumulative Voting (AFHCV), hybrid Gamified method with AHP, FuzzyHCV—hybrid of HCV and Fuzzy Expert System, Interactive Genetic Algorithm (IGA) and search-based method and the combination of PG and AHP. Tonella et al. proposed a search-based approach, called Interactive Genetic Algorithm (IGA), for requirements prioritization [3]. The IGA requirements prioritization approach is a kind of pairwise comparison method that gets the advantages of using a genetic algorithm to reduce the number of pairwise comparisons that need to be elicited from the decision-maker. It merely extracts those pairs of requirements which prevent the ambiguity of similarly ordered or even diversely ordered requirements. This can enhance the scalability of the approach when using for prioritizing a large number of requirements.

A novel technique called Fuzzy Hierarchical Cumulative Voting (FHCV) proposed by [4] and is designed for the prioritization of requirements by combining Fuzzy system with HCV technique. HCV is the base of FHCV. The HCV technique does not take into consideration ambiguity and uncertainty associated with stakeholder's judgment about assigning points to the requirements. Therefore, current HCV seems inadequate for decision-makers when they are not confident about assigning points to requirements. To tackle such problems, a technique known as FHCV has been invented. In FHCV, instead of crisp points/numbers, the use of triangular fuzzy numbers is made. The problem with the FHCV is that the requirements ought to be organized hierarchically in advance, as FHCV does not provide the support for a hierarchical arrangement of requirements.

The proposed Adaptive Fuzzy Hierarchical Cumulative Voting (AFHCV) by [5] uses adaptive mechanisms with existing Fuzzy Hierarchical Cumulative Voting (FHCV) technique, to increase the coverage of events that can occur at runtime. The adaptive device includes Addition of new requirement set, Analysis, and Reallocation of requirements, Assignment and Alteration of priorities and Re-prioritization. The re-prioritization is used to

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improve the results of proposed AFHCV.

Based on the literature, many researchers have found techniques that can reduce effort, time, and costs during requirement prioritization. However, there are still some essential issues neglected by previous researchers. This paper is structured as follows. Section 2 presents the motivation of the study. Section 3 presents the related work, while Section 4 presents the research method with the research question, search process, inclusion, and exclusion criteria and quality assessment. Section 5 describes the research findings, and Section 6 discusses the research findings. Section 7 concludes the paper.

2 MOTIVATION

That requirements have varying importance is a widely acknowledged reality. Techniques for requirements prioritization have not progressed, either theoretically or practically [5]. Based on some reviews, several organizations think that allocating priorities to requirements is an important step, and decisions should be made [5]. Consequently, it seems that few organizations knew about how to allocate priorities or about the way that these priorities should be communicated to the project team members. There are various techniques of requirement prioritization illustrated by researchers in the domain of Software Engineering. Choosing a particular prioritization technique depends upon the type and purpose of the project.

In the last few years, due to the need for flexibility in software projects, Agile Software Development (ASD) methods (e.g., Scrum and Extreme Programming) are becoming more popular [5,6]. ASD gives some benefits compared to traditional software development methods such as the delivery of business value in short iterations, following an incremental and empirical development process.

Additionally, continuous communication (e.g., face to face communication) and a development process open to changes are encouraged by ASD methods. Initially, requirements are defined with customers but are then continuously refined [7].

3 RELATED WORK

This section summarizes related research publications on ASD requirements prioritization. Extensive research has been conducted on the requirements of prioritization techniques. There are various techniques available today for prioritization, each with multiple criteria and pre-conditions.

Different researchers have evaluated the results of varying prioritization approaches, and then based on these results, they have developed new techniques. Karlsson et al. Have conducted a research in which they have evaluated different techniques [8] and recorded the results of different techniques. Karlsson et al. applied the different prioritization methods in order "to prioritize 13 well-defined quality requirements on a small telephony system" [8]. The results provided by most of the techniques, except AHP, are unreliable and prone to errors; on the other hand, AHP is more reliable but time-consuming due to the decisions which one must make in terms of requirements prioritization methods [8]. Though AHP is a reliable technique for requirements prioritization, the technique is complex to implement and time-consuming due to pair comparisons.

The prioritization techniques of Binary Search and Priority Groups are also unreliable because of unsatisfactory results. The role of the consistency index is vital because it helps to reduce human error. All these techniques are applied to only a few requirements, so AHP, even though it is efficient, is still considered problematic when requirements are scaled up [9]. Laurent et al.

have said, "AHP does not scale well because the number of comparisons grows exponentially with the number of requirements" [9].

Research by [10] presented a requirements prioritization approach based on the hierarchical dependencies among requirements. The author provided a new method to reduce the number of comparisons in prioritization techniques such as the AHP method. Furthermore, the author has affirmed that to the best of their knowledge, this is the first approach that targets the dependencies among requirements hierarchies.

A brief survey on the popular prioritization techniques presented by [11], and then introduced their goal-based prioritization technique based on collecting relative weight of requirements concerning goals identified from the stakeholders. Decision factors suggested by [12] like Cost-Benefit to the Organization, Complexity, Requirement Dependencies, and Delivery Date/Schedule are the most important for prioritizing user stories. Understanding the decision factors, project planners would know which requirements should be prioritized to reduce risk and increase stakeholder satisfaction. A methodology where ISO/IEC external metrics have considered was proposed by [13], and the results found an increase in quality of requirements selected, hence influencing the prioritization. Another research on the requirement prioritization process in Agile Development by [14] highlights the three primary aspects: Environment, Process, and Product. Each consists of further factors: Environment has Stakeholder Characteristics, Project Constraints, and Requirement Nature; Process has activities involved in it, and Product has the outcome of the process. It helps to investigate prioritization techniques that will target any of these aspects.

A study on the evaluation of requirement prioritization techniques by [15] found ANP to be the most powerful technique for both dependent as well as independent requirements. The experiment conducted proved that ANP worked even more efficiently for the dependent requirements. A new technique of type absolute scale by improving two methods, which are VOP and Wieger's have been developed through the research done by [16]. They compared three models in their study. A comparison study was conducted to develop a model that includes possible RP factors that significantly affect the RP process. This study was based on the existing RP models from the reviewed literature. The defects and the missing items of the reviewed model were considered when designing the new model.

There are few systematic reviews conducted by the previous researchers in this research domain. As stated by [17], seven prioritization techniques were identified and analyzed. These techniques include; analytic hierarchy process, value-oriented prioritization, cumulative voting, numerical assignment, binary search tree, planning game, and B tree prioritization. A systematic mapping study in software requirements prioritization with specific emphasis on empirical studies was reported by [18] in their research. Most of the studies documented in their work were about the validation of various investigations. Besides, the authors reported the occurrence of studies on techniques and confirmed that there is an infrequent interest in the evaluation of tools that could be beneficial to the industry. Furthermore, the authors, [19] carried out another systematic literature review on software requirements selection and prioritization research with a particular focus on search-based software engineering (SBSE). There were 30 articles that were evaluated with 18 established quality criteria. The review analyzed some quality criteria that must be taken into consideration when prioritizing requirements. A systematic literature review conducted by [20], focuses on the technical aspects only, and it covers issues of scalability, complexity, ease of use, reliability, validation, evolution, and dependencies. This contribution

has missed out the important business aspects and factors. Similarly, many existing latest surveys and current researches by the researchers, on the present work, are not discussed and analyzed. Besides, the authors do not focus merely on requirement prioritization in Agile Software Development (ASD). Therefore, in this systematic literature review, it is focusing more on the issues and limitations in the requirement prioritization, focusing on Agile Software Development. The findings in this study are important as a guideline to the researchers and practitioners who will research this area of study in the future.

4 REVIEW METHOD

The review processes for this SLR followed SLR guidelines for software engineering [21]. Based on the guidelines, there are three primary phases involved in this SLR: planning the review, conducting a review, and reporting the review, as shown in Fig. 1.

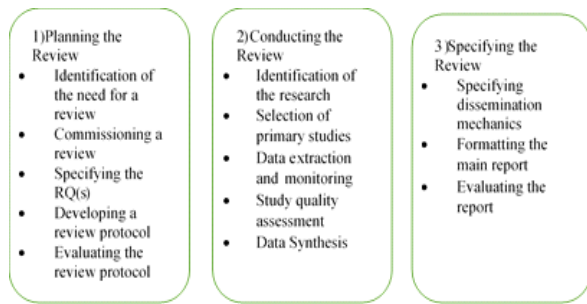


Fig. 1. Systematic Literature Review (SLR) phases and stages

Initially, a set of research questions were formulated based on the objective of this research. The research questions for this SLR include five components. The components are known as PICOC (Population, Intervention, Comparison, Outcomes, and Context). It was proposed by [21]. Table 1 shows the criteria and scope of the research questions.

TABLE 1
CRITERIA AND SCOPE OF RESEARCH QUESTIONS

Criteria	Scope
Population	Challenges, Classification, Analysis and Limitations of requirement prioritization in ASD
Intervention	Requirement prioritization technique in ASD
Comparison	NA
Outcomes	Structure of requirement prioritization the technique applied in ASD
Context	Review(s) of any empirical studies of the requirement prioritization in ASD

This research work aims to identify and analyze existing prioritization techniques and related issues or limitations by illustrating various requirements and prioritization practices, methodologies, frameworks, and models focused on ASD. Four research questions have been defined to achieve the aim of this study:

RQ1: What requirement prioritization techniques used in ASD are available in the research literature?

RQ2: How are the proposed techniques evaluated?

RQ3: What are the issues or limitations in the current ASD

technique?

RQ4: What are the factors involved in RP for ASD?

5 DATA SOURCES

The research questions help to define the limits of the research while exploring the published research in a specified domain. The search process is designed and formulated based on keyword-technical terms derived from existing studies.

Six online databases have been chosen as data sources: ACM Digital Library, Google Scholar, IEEE Xplore Digital Library, ScienceDirect, SpringerLink, and Scopus. The selection of online databases was based on a list available in an online database subscribed to by the University Putra Malaysia's Library under the "Computer Science" subject category.

6 SEARCH PROCESS

Since the search process distinguishes SLR from the traditional survey, it is necessary to follow the search strategy. Trial searches with combinations of terms derived from the research questions need to be done before finalizing the search string. The following steps have been taken to formulate the search query for this SLR:

- Derivation of significant terms from the research question.
- Identifying synonyms of significant terms.
- Identifying keywords in relevant papers or books.
- Boolean OR for alternative synonyms or variants of each keyword.
- Boolean AND to link significant terms.

The researcher attempted several search strings. The researcher would find too many unrelated papers rather than requirement prioritization in agile when using the term "requirement prioritization" alone. Based on that reason and number of trials, the researcher decided to use the term "requirement prioritization" and "agile" linked together both terms with a Boolean 'AND'. Using the terms "requirement prioritization" and "agile," the researcher found almost all papers related to requirement prioritization in agile. The resulting search string is described as follows:

((("Requirement Prioritization" AND "Agile") OR "Prioritization in Agile") OR ("Requirement Prioritization" AND "Agile" AND "classification") OR ("Requirement Prioritization" AND "Agile" AND "challenges") OR ("Requirement Prioritization" AND "Agile" AND "analysis") OR ("Requirement Prioritization" AND "Agile" AND "limitations"))

The researcher performed a manual check of the results of the search string and realized that some of the online databases needed advanced settings, such as IEEE Xplore and Scopus. The researcher then added alternative words and expressions to the search string.

7 STUDY SELECTION

Many research studies were collected, but there was a need to sort out related research studies. The research process was narrowed down to get the most relevant papers, and filtering was performed based on the inclusion and exclusion criteria [22]. Fig. 2 shows the review protocol phases involved in this research.

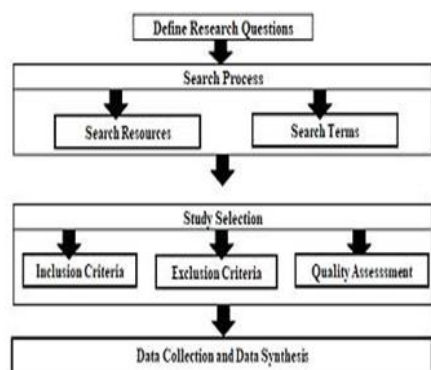


Fig.2. Review protocol phases

7.1 Inclusion and Exclusion Criteria

Inclusion and exclusion criteria for this SLR is based on the research question [21]. Since this SLR is focused on the RP technique in ASD, to select only relevant papers, it was necessary to define inclusion and exclusion criteria. The inclusion and exclusion criteria are shown in Table 2.

TABLE 2
THE INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	Exclusion Criteria
All papers are written in the English language.	Papers are not published in the English language.
Technical aspects of Requirements prioritization was discussed.	Papers that do not have any link with the research questions.
Papers discussing the RP techniques in ASD.	Simple articles which are published on websites.
Papers published on RP in ASD from the year 2013 to 2019.	Papers that are publishes before 2013.
All potential published papers that can answer the research questions.	Duplicate papers.

7.2 Data Extraction and Quality Assessment

The quality assessment for this study covered quantitative and qualitative studies since there were no restrictions in terms of experimental design. A quality assessment study checklist was used to ensure the data extraction process meets the quality criteria [22]. The quality assessment checklist was designed based on the guidelines proposed by [21]. Table 3 shows a list of general questions to measure the quality of the selected studies. Three scales were coded for the quality assessment checklist and given a score: Yes =1; Partially=0.5; No = 0. Based on the item checklist, each article ranged from 0 (very poor) to 5 (very good).

TABLE 3
QUALITY ASSESSMENT CHECKLIST

No	Item	Answer
QAC1	Are the aims and objectives of the research clearly stated?	Yes/No
QAC2	Is the research design clearly specified?	Yes/No/Partially
QAC3	Have the researcher(s) properly carried out the process of data collection?	Yes/No/Partially
QAC4	Have the researcher(s) given enough data to support their results and conclusions?	Yes/No/Partially
QAC5	Is there involved comparison of other technique in the experiment?	Yes/No

8 RESEARCH FINDINGS

In the first stage of the search process, the researcher found 1,301 papers using the defined search terms. After screening titles and abstracts, only 95 papers were potentially relevant. Each of the papers was filtered for referring inclusion and exclusion criteria before being accepted for the synthesis of evidence. At this stage, irrelevant studies and duplicate studies were eliminated. The researcher read the full articles to categorize the paper relevant or not to the research area. Ninety-five papers are very carefully analyzed at the initial stage to extract the required information. Finally, only 25 papers were selected to provide answers to the formulated research questions. Fig. 3 shows the results of the search and selection papers process.

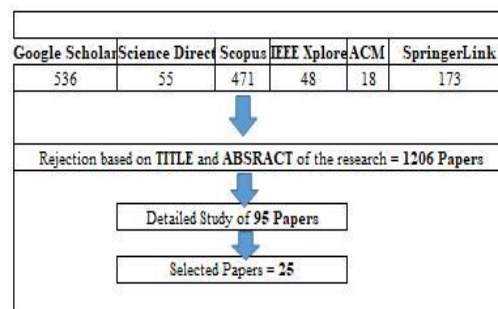


Fig. 3. Search and Selection Papers Stage

8.1 Quality of Factors

Table 4 indicates the quality assessment scores for the final 25 identified papers. Six studies (24 percent) were fair, nine studies (36 percent) were good, and ten studies (40 percent) were of outstanding quality. There was no paper rated as poor quality. Therefore, all selected papers were included in the next phase for further analysis.

TABLE 4
QUALITY ASSESSMENT SCORES

Quality Scale	Very Poor (>=1)	Poor (>=2)	Fair (>=3)	Good (>=4)	Very Good (=5)	Total
No of Studies	0	0	6	9	10	25
Percentage (%)	0	0	24	36	40	100

9 DISCUSSION

This section presents and discusses the results related to the research questions. A detailed description of the findings is presented in this section with the research questions.

RQ1: What requirement prioritization techniques used in ASD are available in the research literature?

The purpose of this research question is to investigate the current requirement prioritization techniques used in ASD. From the SLR studies, a majority of the papers (12 percent) implemented the Analytic Hierarchy Process (AHP) technique during the RP process. Some of the papers combined techniques by using different types of hybrid techniques, such as Adaptive Fuzzy Hierarchical Cumulative Voting (AFHCV), hybrid Gamified method with AHP, FuzzyHCV –hybrid of HCV and Fuzzy Expert System, Interactive Genetic Algorithm (IGA) and search-based method and the combination of PG and AHP. From the literature, some researchers agreed that using more than one RP technique during the prioritization process can give better results and maybe more efficient [5, 6, 8, 16]. Table 5 depicts the most commonly utilized techniques based on SLR evaluations.

TABLE 5
MOST POPULAR AND UTILIZED TECHNIQUES

Techniques	Authors
Analytic Hierarchy Process (AHP)	[8,15,17,23,24,27]
Cumulative Voting (CV)	[23,5,25,24,26]
Kano Model	[13,24]
Value-Oriented Prioritization	[15,16]

Even though the AHP and CV methods proved to be a very reliable and widely utilized technique among researchers, it still suffers from scalability problems and is time-consuming, as more comparisons are needed when the number of requirements increases [14,15,20]. There is a need for AHP and CV techniques, to combine or hybrid with other methods to provide more reliable and effective results to overcome the scalability and time-consuming issues.

RQ2: How are the proposed techniques evaluated?

In accessing the effectiveness and efficiency of the existing techniques, the reporting of empirical studies is very important. Most of the papers that used a case study agreed to apply real software system or industry for their future work [10,14,30,31]. Out of 25 papers, 18 reported that they used a case study to evaluate their proposed techniques. The remaining seven papers said that they used online data for their experiment.

RQ3: What are the issues or limitations in the current ASD technique?

Many techniques have been developed by numerous researchers. However, the existing RP techniques failed to resolve some issues which are considered as limitations to their research. The limitations will be used as a basis to improve the RP techniques in ASD. In summary, Table 6 depicts the significant issues or limitations of existing RP techniques in ASD.

TABLE 6
SIGNIFICANT ISSUES OR LIMITATIONS OF RP TECHNIQUES IN ASD

Issues/Limitations	Authors
Existing techniques suffer from severe limitations in terms of scalability, complexity, uncertainty, time Consumption	[32,33,34,28,11,20]
Issue starvation (two requirements with The same priority)	[34]
Limitations in term of the dependencies among Requirements	[10,30,11,35,36]
Limited research focused on the non-functional requirements	[37,24,28,38,39]
Less automation approach	[32,37,34,24]

There are few challenges of RP techniques in Agile Software Development. Table 7 represents the significant challenges of RP techniques focusing on ASD. The challenges facing by the practitioners and researchers during the prioritization process are such as conflict between the stakeholders, changes done by the stakeholders in the priority list that can lead towards rework or re-prioritization and the selection of factors involved during the RP process.

TABLE 7
SIGNIFICANT CHALLENGES OF RP TECHNIQUES IN ASD

Challenges	Authors
A conflict between the Stakeholders	[29,24,11,3,16,10,28]
Changes in the priority list lead towards rework	[32,16,40]
Requirement selection factors during RP process	[28,6,7,5]

RQ4: What are the factors involved in RP in ASD?

The main goal of ASD is to deliver business value (Agile Manifesto). One way to achieve this is “the continuous and value-driven requirements prioritization from customer’s respective” [40,18]. Factors are aspects that are taken into consideration while prioritizing requirements. It is easy to prioritize using a single factor. However, when considering multiple factors, the high priority ranked requirements would receive less priority with some change in factors [9,11]. Through the SLR done by the researchers, there are some factors involved during RP in ASD. All factors depend on the purpose of the research since not all factors are suitable to be used during the RP process. Practitioners need to select the factors according to their project. Table 8 shows some factors that are typically implemented in the current RP process.

TABLE 8
SUMMARY OF RP USING DIFFERENT FACTORS

References	Factors										
	Cost	Value	Dependencies	Risk	Volatility	Importance	Decision-making	Scalability	Time	Complexity	Effort
Alawneh [2018]	/	/	/	/							
Viswanathan, Nair, & Krishnan [2017]		/	/	/		/			/		/
Rahim et al. [2017]	/	/		/	/				/		
Silva, Ramos, & Silva [2017]		/	/							/	
Kolondinos & Ginz [2017]											/
Batra & Bhattacharya [2016]	/			/			/	/		/	/
Hujainah et al. [2016]	/		/	/		/	/	/	/	/	/
Al-Ta'ani & Razali [2016]	/	/	/						/	/	
Hudda et al. [2016]						/				/	
Scheerer, Bick, Hidenbrand, & Henzl [2015]			/								
Sher, Jawawi, Mohamad, & Babar [2014]								/	/	/	
Daneva et al. [2013]		/	/	/							
Kukreja, Swaroop, Boehm, & Padmanabhuni [2013]		/									
Al-Ta'ani & Razali [2013]											/

Based on the results stated in Table 8, the most implemented factor by the researchers is the ‘value’ factor, followed by the ‘risk’ factor. Nevertheless, according to [41], in the future, various other factors essential for users, developers, and testers can be considered for prioritization of the requirements. These new methods can be included in the prioritization process in ASD for the rapid and high-quality delivery of software products to customers.

10 CONCLUSION

The main objective of this SLR was to identify the RP techniques used in ASD and the current issues related to the RP techniques. The analysis in this SLR has been based on the constructed research question. A total of 25 primary studies were selected and analyzed. The results of this SLR show that many researchers have not reported their empirical results thoroughly. However, the results have shown that RP in ASD research continues to increase annually as the RP in the ASD technique is increasingly essential [16,32,41]. The essence of this SLR may be used to identify areas for possible improvement for RP in ASD. A systematic literature review has shown that researchers have a deep interest in the area of RP, especially in ASD. More research areas were introduced after studying the published research work, and several further improvements are desirable. As mentioned in

Table 6 and Table 7, there are still some issues, limitations, and challenges for RP in ASD that can be taken into consideration in future works.

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